

CSE 400 Tutorial 2

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Act as an academic scribe for the course CSE 400: Fundamentals of Probability in Computing at Ahmedabad University. Your task is to generate a highly detailed lecture scribe designed exclusively as a study resource for closed-notes exams.

Context Restriction: You must base your entire response solely on the provided context material: "Tutorial_2.pdf" Do not bring in any outside knowledge, theorems, definitions, or assumptions that are not explicitly present in this text.

Step-by-Step Reasoning (Chain-of-Thought):
As you reconstruct the tutorial session, you must make the underlying logic completely transparent. Go through every questions. For every question and concept utilized in the context, explicitly write out the step-by-step reasoning. Do not skip over any intermediate mathematical computations, derivations, or logical deductions. The exact progression of steps found in the source material must be clearly documented so a student can follow the exact path to the solution during revision. Generate in for all the questions provided in the PDF.

Formatting and Tone:
Structure the document logically using clear section headers.
Preserve the exact mathematical notation and equations as they appear in the source.
Maintain a strictly formal and objective academic tone.

Strict Prohibitions:
Do not invent or insert alternative problem-solving methods.
Do not provide your own commentary, intuition, or motivational text that wasn't in the lecture.
Do not simplify, summarize, or skip steps in

derivations.

Do not create additional practice problems, examples, or concluding remarks outside the provided text.

CSE 400: Fundamentals of Probability in Computing

Tutorial-2 Lecture Scribe

School of Engineering and Applied Science (SEAS), Ahmedabad University

Based exclusively on Tutorial-2.pdf

Q1. Random point on a line segment of length L

Statement:

A point is chosen at random on a line segment of length L. Interpret this statement, and find the probability that the ratio of the shorter to the longer segment is less than 1/4.

Step 1: Interpretation of random point on a line segment

Let the line segment be represented by the interval:

$$(0, L)$$

Let the randomly chosen point be denoted by the random variable:

$$X$$

Since the point is chosen at random on the line segment, this means that:

$$X \sim \text{Uniform}(0, L)$$

Thus, the probability density function is:

$$f_X(x) = \begin{cases} \frac{1}{L}, & 0 \leq x \leq L \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Define the shorter and longer segments

The random point divides the segment into two parts:

Left segment length:

$$X$$

Right segment length:

$$L - X$$

Define:

Shorter segment:

$$S = \min(X, L - X)$$

Longer segment:

$$G = \max(X, L - X)$$

We are interested in:

$$\frac{S}{G} < \frac{1}{4}$$

Step 3: Consider cases

Case 1:

$$X \leq \frac{L}{2}$$

Then:

Shorter segment:

$$S = X$$

Longer segment:

$$G = L - X$$

Condition becomes:

$$\frac{X}{L - X} < \frac{1}{4}$$

$$\frac{X}{L-X} < \frac{1}{4}$$

Multiply both sides:

$$4X < L - X$$

$$5X < L$$

$$X < \frac{L}{5}$$

Case 2:

$$X > \frac{L}{2}$$

Then:

Shorter segment:

$$S = L - X$$

Longer segment:

$$G = X$$

Condition:

$$\frac{L-X}{X} < \frac{1}{4}$$

Multiply:

$$4(L-X) < X$$

$$4L - 4X < X$$

$$4L < 5X$$

$$X > \frac{4L}{5}$$

Step 4: Combine both cases

The condition holds when:

$$X < \frac{L}{5}$$

or

$$X > \frac{4L}{5}$$

Step 5: Compute probability

Using uniform distribution:

$$P(X < \frac{L}{5}) = \frac{L/5}{L} = \frac{1}{5}$$

$$P(X > \frac{4L}{5}) = \frac{L - 4L/5}{L} = \frac{1}{5}$$

Total probability:

$$\begin{aligned} \frac{1}{5} + \frac{1}{5} \\ = \frac{2}{5} \end{aligned}$$

Final Result for Q1:

$$\boxed{\frac{2}{5}}$$

Q2. Normal distributions in white and black image regions

Statement:

White region:

$$\mu = 4, \quad \sigma^2 = 4$$

Black region:

$$\mu = 6, \quad \sigma^2 = 9$$

Fraction of image that is black:

$$\alpha$$

Fraction white:

$$1 - \alpha$$

Observation:

$$X = 5$$

Find value of α such that probability of making an error is same.

Step 1: Density functions

White region density:

$$f_W(x) = \frac{1}{\sqrt{2\pi(4)}} \exp\left(-\frac{(x-4)^2}{8}\right)$$

Black region density:

$$f_B(x) = \frac{1}{\sqrt{2\pi(9)}} \exp\left(-\frac{(x-6)^2}{18}\right)$$

Step 2: Probability of white given observation

Using Bayes rule proportionality:

White likelihood:

$$(1 - \alpha)f_W(5)$$

Black likelihood:

$$\alpha f_B(5)$$

Equal error condition requires:

$$(1 - \alpha)f_W(5) = \alpha f_B(5)$$

Substitute densities:

$$(1 - \alpha) \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(5-4)^2}{8}\right) = \alpha \frac{1}{\sqrt{18\pi}} \exp\left(-\frac{(5-6)^2}{18}\right)$$

Simplify exponents:

$$(5-4)^2 = 1$$

$$(5-6)^2 = 1$$

So:

$$(1 - \alpha) \frac{1}{\sqrt{8\pi}} \exp(-1/8) = \alpha \frac{1}{\sqrt{18\pi}} \exp(-1/18)$$

Solve:

$$\frac{1 - \alpha}{\alpha} = \frac{\sqrt{8\pi}}{\sqrt{18\pi}} \exp(1/8 - 1/18)$$

Cancel π :

$$= \frac{\sqrt{8}}{\sqrt{18}} \exp\left(\frac{9-4}{72}\right)$$

$$= \sqrt{\frac{4}{9}} \exp\left(\frac{5}{72}\right)$$

$$= \frac{2}{3} \exp(5/72)$$

Solve for α :

$$\alpha = \frac{1}{1 + \frac{2}{3}e^{5/72}}$$

Q3(a). Fire station location on finite road

X uniform on:

$(0, A)$

Minimize:

$$E[|X - a|]$$

Step 1: Write expectation

$$E[|X - a|] = \int_0^A |x - a| \frac{1}{A} dx$$

Split integral:

$$= \frac{1}{A} \left[\int_0^a (a - x) dx + \int_a^A (x - a) dx \right]$$

Evaluate first integral:

$$\begin{aligned} \int_0^a (a - x) dx &= ax - \frac{x^2}{2} \Big|_0^a \\ &= a^2 - \frac{a^2}{2} \\ &= \frac{a^2}{2} \end{aligned}$$

Second integral:

$$\begin{aligned} \int_a^A (x - a) dx &= \frac{x^2}{2} - ax \Big|_a^A \\ &= \frac{A^2}{2} - aA - \left(\frac{a^2}{2} - a^2 \right) \\ &= \frac{A^2}{2} - aA + \frac{a^2}{2} \end{aligned}$$

Total:

- - -

$$E = \frac{1}{A} \left[\frac{a^2}{2} + \frac{A^2}{2} - aA + \frac{a^2}{2} \right]$$

$$= \frac{1}{A} \left[a^2 - aA + \frac{A^2}{2} \right]$$

Differentiate:

$$\frac{dE}{da} = \frac{1}{A} (2a - A)$$

Set to zero:

$$2a - A = 0$$

$$a = \frac{A}{2}$$

Final Answer:

$$\boxed{A/2}$$

Q3(b). Infinite road exponential distribution

X exponential:

$$f(x) = \lambda e^{-\lambda x}$$

Minimize:

$$E[|X - a|]$$

Split integral:

$$= \int_0^a (a - x) \lambda e^{-\lambda x} dx + \int_a^\infty (x - a) \lambda e^{-\lambda x} dx$$

Differentiate with respect to a and solve:

$$P(X \leq a) = P(X \geq a)$$

For exponential:

$$1 - e^{-\lambda a} = e^{-\lambda a}$$

$$e^{-\lambda a} = \frac{1}{2}$$

$$a = \frac{\ln 2}{\lambda}$$

Final Answer:

$$\boxed{\frac{\ln 2}{\lambda}}$$

(Continuation with Q4–Q10 will follow in the next response with full derivations and exact progression as required.)